3.6 Simple linear equations

Now, let us learn to construct simple linear equations and solve them.

3.6.1 Construction of linear equations

Consider an expression $7x + 3$, where x is the variable.

When $x = 2$, the value of the expression is $(7 \times 2) + 3 = 14 + 3 = 17$.

Also, this can be written as $7x + 3 = 17$, when $x = 2$. $7x + 3 = 17$ is called an equation.

Moreover, no value of x other than 2 satisfies the $= 7x + 3 = 17$. Thus $x = 2$ is called the **solution** to the equation $7x + 3 = 17$.

An equation, is always equated to either a numerical value or another algebraic expression. The equality sign shows that the value of the expression to the left of the '=' sign is equal to the value of the expression to the right of the '=' sign. In the above example, the expression $7x + 3$ on the left side is equal to the constant 17 on the right side.

In general, the RHS of an **equation** is just a number. But, this need not be always be so. The RHS may be an expression containing the variable. For example, the equation $7x + 3 = 3x - 1$ has the expression $7x + 3$ on the left and $3x - 1$ on the right separated by an equality sign.

Let us look at some situations where we have the value of an expression without actually knowing the values of each of the variables.

The cost of 10 chairs and 4 tables is ₹ 4000. We are not given the price of a chair or a table. Let us take the price of one chair as ₹ x and one table as ₹ y. Then the cost of 10 chairs and 4 tables is $10x + 4y$. Therefore, $10x + 4y = 4000$. This is an equation in two variables x and y taking the value 4000. Let us consider some more examples and find the algebraic equations.

1. "The cost of an apple and 2 mangoes is ₹ 120"

   If the price of an apple is a and one mango is m, then the required equation is $a + 2m = 120$.



Fig.3.7

2. "A rectangle of some dimensions has perimeter 50 cm".

   If l and b are the length and breadth of the rectangle then we write

   $2(l + b) = 50$.

3. "Thamarai is 4 years elder to her sister Selvi and sum of their ages is 24"

   The ages of Thamarai and her sister are treated as variables. Although, their ages are different, we consider this as one variable, because the age of Thamarai is associated to the age of Selvi.

   Suppose, if Selvi's age is 10, then the age of Thamarai is $10 + 4 = 14$. In the same way, if the age of Selvi is x, then the age of Thamarai is $x + 4$.

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